Supplementary Information for "Resistance-Aware Computational Analysis of Antimalarial Leads from African Natural Products"

August 9, 2026

Table 1: Docking protocol validation: enrichment metrics and re-docking accuracy for the four wild-type targets. Values reproduce the corrected benchmarks from the parent study (V2-corrected grids; AutoDock Vina + DiffDock consensus). MMV Malaria Box values are consensus-score ROC-AUCs; DEKOIS 2.0 validation used AutoDock Vina only. BEDROC values use $\alpha = 20$ (MMV). Re-docking RMSD is the heavy-atom RMSD between the top-scoring docked pose and the co-crystallised ligand; MTX redocking failed (RMSD ≈ 30 Å).

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC
PfDHFR (7F3Y)	Vina redock	–	–	–	–
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.309
	DEKOIS (Vina only)	0.509	–	0.51	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.434
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810
PfClpR (4GM2; not PfClpP)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810

Identity note.—PDB 4GM2 is PfClpR rather than PfClpP; the historical 4GM2 docking record is therefore not PfClpP validation.

Note.—Redocking against V2-corrected grids gave 100% success for the five alignable co-crystallised ligands. MTX (PfDHFR co-crystallised ligand) could not be redocked and is reported as N/A. AutoDock Vina + Gnina consensus scoring was used in the present validation study; the enrichment metrics above are taken from the parent study (Vina + DiffDock) to ensure consistency with the corrected P1 data.

Table 2: Endpoint-analysis interpretation for the four parent-study consensus complexes used for production MD. These systems are distinct from the 17 set-C polypharmacology candidates. Mean ± standard deviation is shown only for the single interpretable MM-GBSA result from 101 snapshots; dissociated or conversion-corrupted systems are reported as N/A. Mutant RRS is a docking-based analysis of set C and is not assigned to these MD systems.

Ligand	Target	ΔG_{bind}	Std. dev.	Interpretation
164	PfClpR-labelled cohort*-WT	–	–	N/A; ligand dissociated‡
201	PfDHFR-WT	–	–	N/A; ligand dissociated‡
214	PfCRT-WT	−18.25	0.40	Interpretable (gmx_MMPBSA)
438	PfATP4-WT	–	–	N/A; conversion corrupted§

*PDB 4GM2 is PfClpR rather than PfClpP; the 164-labelled system is therefore not structural validation of

‡Ligand dissociated during production; no binding free energy is inferred.

§Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion.

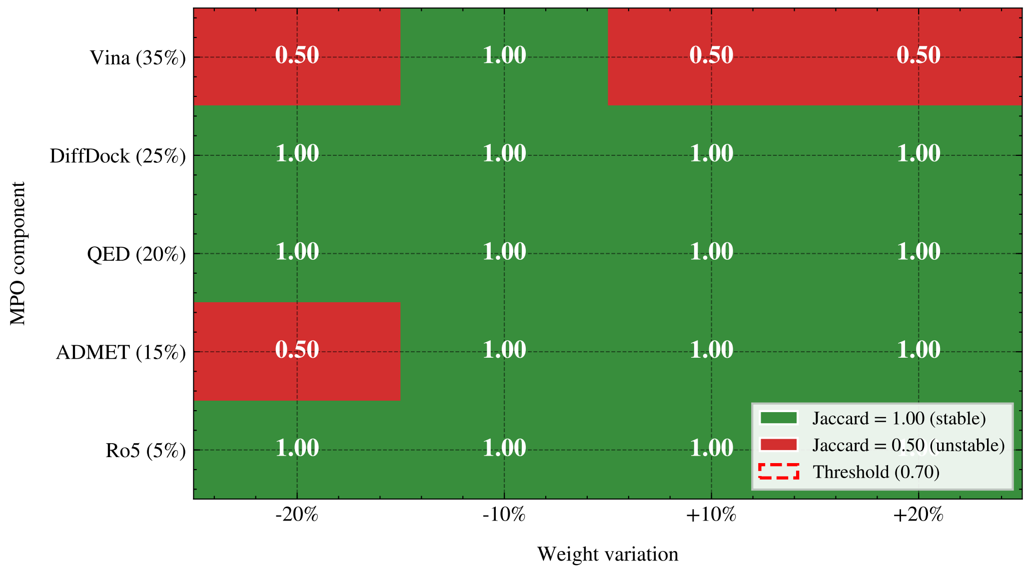

Figure 1: MPO sensitivity analysis heatmap. Each cell shows the Jaccard similarity of the top-20 hit list when the corresponding MPO weight is varied by ±20 %. Green (1.00) indicates the hit list is unchanged; red (0.50) indicates substantial reordering.

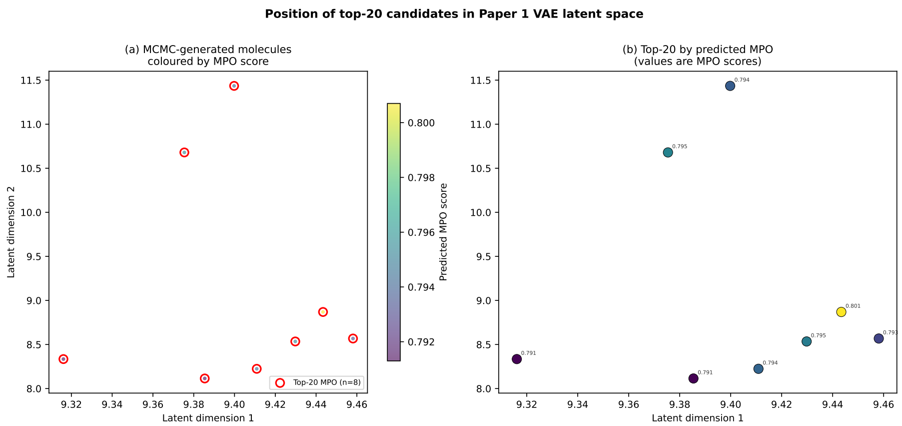

Figure 2: Position of the top-20 candidates in the Paper 1 VAE latent space. Molecules are coloured by predicted MPO score; top-20 by MPO are highlighted in red.

Table 3: ADMET cross-validation schema for the 17 set-C candidates and 5 reference controls. The comparison would use ADMET-AI, ADMETlab 3.0, and SwissADME, with Spearman correlations and inter-tool disagreement flags; no numerical comparison is reported here.

Compound	ADMET-AI LogS	ADMETlab LogS	SwissADME LogS	hERG pIC_{50}	Flagged?
No numerical three-tool ADMET comparison is reported here.					

Table 4: ACSI weight sensitivity analysis. Each of the four ACSI weights (w_{DrugBank}, w_{ANPDB}, w_{fsp^3}, w_{NPL}) was varied by ±20 % while keeping the remaining weights proportional. The Jaccard index measures overlap of the resulting top-20 ACSI-ranked list with the baseline ranking. A Jaccard ≥ 0.70 (14/20 compounds stable) indicates robustness.

Weight varied	Variation (%)	Mean ACSI (top-20)	Std. dev.	Jaccard vs. baseline
Data to be populated after ACSI computation.				

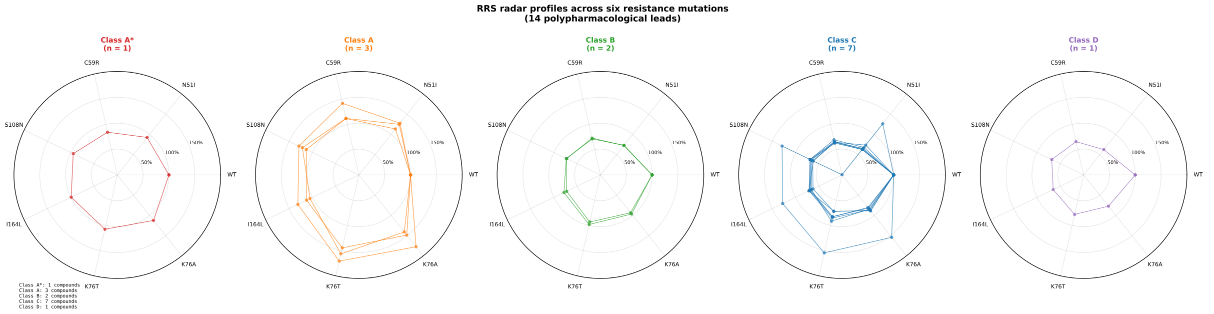

Figure 3: Per-compound, docking-derived RRS profiles across all six resistance mutations for the 17 set-C polypharmacological leads. The four parent-study MD systems are not included. Each axis represents one mutation; concentric rings indicate 50 %, 100 %, and 150 % RRS (WT-normalised). Class A* and Class A compounds fill the outer region (RRS > 100 %).

Additional MD assessment. A diagnostic continuation of the parent PfDHFR and PfCRT systems did not yield a usable new trajectory. It therefore contributed no new MM-GBSA estimate or MD-RRS result. No candidate-specific Set-C production MD was obtained.